SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science
COURSE CODE: DSA0401

COURSE OUTCOME COVERED

CO1: Analyze *the* role of data science, facets *of* data, *and the big data ecosystem for informed* decision-making. (BL2)

QUESTIONS: 20

Annexure A

SHORT ANSWER TEST
Total Marks:100

Code of Conduct:

IskAzeem

(192472307)

(Name / *Reg No)* certify *that this submission is my original work and that* I *have adhered to the guidelines specified for this assessment*.

I understand that any violation of academic *integrity rules will result in disciplinary action*.

Signature of the student*:*

Annexure A

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?

2. A hospital has patient records for many years. How can Data Science be applied?

3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

4. A bank wants to detect suspicious transactions. How can Data Science be useful? 5. A company wants to reduce customer loss. How can Data Science support this?

6. A school wants to improve student performance. Mention one benefit of Data Science in this case.

7. A supermarket wants to know which products sell more during festivals. How can Data Science help?

8. A transport department wants to reduce traffic congestion. How can Data Science be applied?

9. A farmer wants to improve crop yield. Give one use of Data Science.

10. A company has data in Excel tables, customer reviews, images, and videos. Which

facet of data is shown here?

11. A weather app updates temperature every minute. Which facet of data is involved? 12. A university stores student details in rows and columns. What type of data structure is

this?

13. Customer comments on social media are collected for analysis. What type of data is

this?

14. A company collects a very large amount of daily user activity data from its mobile

app. Which Big Data characteristic is applied?

15. A hospital dataset contains missing age values and incorrect phone numbers. Which

data quality issue is present?

16. A business wants to convert customer data into useful sales decisions. Which Big Data

characteristic is important?

17. A company stores large files in Hadoop Distributed File System. Which part of the

Big Data ecosystem is used?

18. A company wants to process large datasets using parallel computation. Which Big

Data tool can be used?

19. An organization wants to collect live sensor data from IoT

devices. Which Big Data

ecosystem component is suitable?

20. A company wants to visualize Big Data results for decision-makers. Which tool or

method can be applied?

RUBRICS FOR CALCULATION

Criteria

Weightage

Level 4- Exemplary (S Marks)

Level 3- Proficient (4 Marks)

Level 2 – Developing (2-3 Marks)

Demonstrates clear

Good

Conceptual Understanding

and complete

30

understanding

nderstanding of the

with minor gaps

Basic understanding with some misconceptions

Level 1 – Beginning (0-1 Mark)

Poor or incorrect

understanding

concept

Accuracy of

Answer
25
25

Correctly applies
Application of
20
ncepts to solve the
with minor

Knowledge
question problem
istakes

Completeness of
Covers all required
15
Response
arts of the question

Clear, concise, and
Klarity &
10
Presentation
well-organized
with minor issues poorly structured
Completely correct Mostly correct
answer with no errors with minor errors noticeable errors

Applies conceptsLimited or partially

Covers most parts Covers only some with miner

Understandable Somewhat unclear or Difficult to
Partially correct with

Incorrect or relevant

nswer

correct application
No or incorrect application

ncomplete or

parts
missing response

understand

06/10/2026/

Sk. Azeem 192472307

1) Data

science can Data science and Past

Performance
to
can analyze attendance, marks, assignments
Predict student at risk

of failing.

2) Data
science can
Can analyze Patient history
to predict diseases, support
diagnosis, and improve
treatment decisions.

3) This is Recommendation system using Data science to suggest product based Previous purchases.

4) Data science on to customer interests and

can use fraud detection techniques to identify unusual transaction

Patterns and detect possible

fraud.

5) Data Science can perform customer churn prediction to identify customers likely to leave and help the

6)

company

one

каке

Preventive actions

benefit is early identification of weak students,

allowing teachers to provide additional supports.

ко

7) Data science can analyze sales Patterns and customer purchasing behavior to

that

ident

identify Products

sell

More

during festivals.

8)

Data science can

vehicle movement

routes

movement

(x) signals.

analyze traffic data, GPS data,

and to predict congestion
to predict congestion and
optimize

analyze weather, soil, and crop
data
best time for planting and
9) Data Science can
analyze
recommend the
उ
no

Irrigation.

10) This shows the variety

in different formats such

variety of data, meaning dara exists

as Structured, semi-

Structured, and unstructured data.